cloudSmoker2 PCB Bring-Up Procedure (v2)

Board: cloudSmoker_ver2f (schematic rev 2f, 2026-06-16) Source: schematic PDF, V2 Micontroller Pin Out wiki page, owner comments. Bench: lab PSU (current limited) for first power-up, then Nitecore power bank. Rigol DS1054Z scope.

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1. Board Architecture

1.1 Power

- **Input:** Nitecore 5V power bank → USB-C U3 (VBUS). CC1/CC2 each have 5.1k (R5, R7) to GND, which presents a 5V sink.
- **5V rail (+5V):** Thermistor bias (PR1–PR4, 120k 0.1%), ADS1115 (U6) VDD, display module, level-shifter 5V side, shunt R8 (100 mΩ) in series with the input.
- **3.3V rail (+3V3):** LDO U1 (TLV76733DGNR) from +5V. Feeds ESP32-C3 (U4), INA226 (U2), LED1, LED2, LED3, and the level shifter 3.3V side. Encoder module is powered from this rail (J8/J7 pin 5). Display logic is 3.3V.
- LDO (U1 TLV76733): EN is pin 5, tied to IN (+5V). OUT is pin 1.
- Decoupling: C1 1µF and C2 2.2µF on the LDO; C3 100nF on the 3.3V rail near INA226; C6 100nF on VBUSF; C9 1µF on the RESET switch; C7 100nF and C8 10µF on the 3.3V rail near the ESP32; C4, C5 470nF on VBUS.

1.2 ESP32-C3 pin map (U4, from pin-out wiki)

Pin	Name	Net	Connects to / note
1	3V3	+3V3	LDO output, local decoupling
2	EN	RESET	SW3 reset, R17 10k pull-up, TP6 (TP_EN)
3	IO4	DISP_DC	Display header
4	IO5	DISP_BL	Display backlight, PWM (logic only)
5	IO6	SPI_SCK	Display. TP7 (TP_SPI_SCK) on bottom, near pin
6	IO7	SPI_MOSI	Display
7	IO8	DISP_CS	Display. Strapping pin, high at boot is safe
8	IO9	BOOT_DISP_RST	SW2 boot, R16 10k pull-up, TP8 (TP_GIO9)
9	GND	GND	Thermal vias to ground plane
10	IO10	ENC_SW	Encoder switch
11	IO20 / RXD	DEBUG_RX	Debug UART header
12	IO21 / TXD	DEBUG_TX/LED	Debug UART header and LED3 via R9
13	IO18	USB_D-	U5 (SRV05) to U3, 90 Ω diff pair
14	IO19	USB_D+	U5 (SRV05) to U3, 90 Ω diff pair
15	IO3	ENC_CLK	Encoder
16	IO2	ENC_DT	Encoder
17	IO1	I2C_SCL_3V3	Q2 (BSS138) to I2C_SCL_5V
18	IO0	I2C_SDA_3V3	Q1 (BSS138) to I2C_SDA_5V

1.3 I2C topology

Device	Address	Supply / bus	Notes
ADS1115 (U6)	**0x48**	5V bus (I2C_SDA_5V, I2C_SCL_5V)	ADDR tied to GND. Pull-ups R19, R21 (10k nominal) to +5V
INA226 (U2)	**0x40**	3.3V bus (I2C_SDA_3V3, I2C_SCL_3V3)	VS = +3V3. Pull-ups R18, R20 (5.1k nominal) to +3V3
Level shifter	n/a	Q1 (SDA), Q2 (SCL), both BSS138	Open-drain shifter between the 3.3V and 5V buses

Assembly note: nine 10k resistors were substituted with 9.1k (0603). Confirm the value of R19 and R21 on your board.

1.4 Thermistor front-end (per channel)

- Jack signal to a 120k bias to +5V (PR1–PR4), with the same node going to the ADS1115 input.
- Jack pins: 1 = sleeve, 2 = tip (signal), 3 = tip switch. Off-board, pin 1 and pin 3 are wired together and, with no plug, all three pins are continuous. That node goes to PCB GND, so the signal net is shorted to GND when empty. Inserting a probe breaks the internal connection and pin 2 carries the thermistor signal.
- There is no separate detect net. The PCB sees only GND and signal per jack. Presence is detected from the ADC reading (near 0 V when empty, a divider voltage when a probe is in).
- Series R22–R25 (10k, fitted 9.1k) and shunt C10–C13 (1µF) form the low-pass filter at about 17.5 Hz (design 16 Hz).
- Route: jack → U7 TVS → filter → ADS1115 AIN0 (PIT), AIN1 (MEAT1), AIN2 (MEAT2), AIN3 (MEAT3).

1.5 Test points

TP	Net	Location
TP1	GND clip	Board edge, bottom layer
TP2	GND	
TP3	+3V3	
TP4	GND	
TP5	+5V	
TP6	RESET (EN)	~3.3V expected
TP7	SPI_SCK	Near IO6, keep scope lead short
TP8	IO9 (BOOT)	
TP9	GND	
TP10	PIT_IN	Bottom layer
TP11	MEAT1_IN	Bottom layer
TP12	AGND	
TP13	MEAT2_IN	Bottom layer
TP14	MEAT3_IN	Bottom layer

Test points are on the bottom layer, except TP7. Use a flip, a pogo jig, or a ground clip on TP1.

1.6 LEDs

- **LED1 (red, R1 2.2k)** and **LED2 (white, R2 1k)**: power indicators on +3V3. Solder bridges JP1 and JP2 can be cut to remove LED draw.
- **LED3 (yellow, R9 100 Ω)**: on IO21 (TXD). Firmware-driven, flickers with UART traffic.

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2. Pre-Power Checks (PSU and power bank disconnected)

Use the DMM in the 20 kΩ range unless noted. Board unpowered.

2.1 Power rails

1. **+5V (TP5) to GND (TP1):** expect several kΩ or more, ideally >10 kΩ. Below 1 kΩ: **stop**.
2. **+3V3 (TP3) to GND:** expect several kΩ or more. Below 1 kΩ: **stop**.
3. **U3 VBUS to GND:** same as TP5 to GND. Confirms the USB-C entry is not shorted.

2.2 I2C pull-ups (verify the resistor value on the board)

1. **+3V3 to I2C_SDA_3V3** (R18): expect ~5.1k.
2. **+3V3 to I2C_SCL_3V3** (R20): expect ~5.1k.
3. **+5V to I2C_SDA_5V** (R19): expect ~10k (9.1k if substituted).
4. **+5V to I2C_SCL_5V** (R21): expect ~10k (9.1k if substituted).
5. **Cross-bus isolation:** I2C_SDA_3V3 to I2C_SCL_3V3 should be open. I2C_SDA_5V to I2C_SCL_5V should be open. Between the 3.3V and 5V sides of the same line, expect high resistance except through the BSS138 body diode.

2.3 Level shifter (Q1, Q2, BSS138)

1. Q1 and Q2 (BSS138), confirmed: gate pin 1 (to +3V3), source pin 2 (3.3V-side net), drain pin 3 (5V-side net). In diode mode, the body diode should conduct in one direction only, from the source side to the drain side.
2. Confirm that gate, source and drain on Q1 and Q2 are not shorted to ground or to each other.

2.4 Thermistor jacks (no plugs)

1. For each jack, with no plug, measure the signal net (PIT_IN, MEAT1_IN, MEAT2_IN, MEAT3_IN) to GND: expect about 0 Ω, since the jack wire shorts the signal to GND. Measure the jack pins directly: pins 1, 2 and 3 should all be continuous. If a channel is open with no plug, the jack wire or PCB GND connection is missing.

2.5 ESP32

1. RESET (TP6) to GND with the board unpowered: expect a high resistance (R17 holds the line up when powered). A low resistance means a short on EN.

Go condition for Phase 3: all of the above in range, no shorts.

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3. First Power-On (lab PSU, current limited)

3.1 Setup

- Lab PSU: **5.0 V**, current limit **250 mA**, output off.
- Connect + to the USB-C VBUS pad on U3, and − to a GND pad (TP1 clip).
- Do not connect the display, encoder, or thermistor jacks.

3.2 Power-up

- Turn output on at 0 V, then raise the voltage over about 5 s.
- Watch the ammeter throughout.
- Stop immediately if the current reaches the 250 mA limit, or exceeds about 150 mA during the ramp.

3.3 Expected behaviour

- Current draw stabilises at roughly 20 to 60 mA with the ESP32 running and no firmware loaded.
- **LED1 (red) and LED2 (white)** glow.
- LED3 may be on steady or off, depending on what the ESP32 is doing at boot.
- No smell, no buzzing, no visible smoke.
- Board is warm but not hot after a minute.

3.4 Immediate DC checks (DMM)

- TP5 (+5V): 4.9 to 5.1 V.
- TP3 (+3V3): 3.2 to 3.4 V.
- TP6 (RESET/EN): about 3.3 V.
- TP2 and TP9 (GND): 0 V relative to TP1.

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4. Power Rail Quality (scope, Rigol DS1054Z)

4.1 +5V ripple (TP5)

- DC coupling, 1X probe, 100 mV/div, 500 µs/div, Ch1.
- Expect <100 mV peak-to-peak.

4.2 +3V3 ripple (TP3)

- DC coupling, 1X probe, 50 mV/div, 500 µs/div, Ch1.
- Expect <50 mV peak-to-peak.
- Above 100 mV: check C1, C2 (LDO) and C7, C8 (MCU).

4.3 Power-on ramp

- Trigger on rising edge of TP5 at 2V.
- Timebase 1 ms/div. Capture the ramp of TP5 and TP3 together.
- Look for overshoot or ringing on TP3 as the LDO comes up. Expect none.

4.4 Load step (optional, after firmware)

- Trigger on TP3 dropping by 100 mV.
- Expect droop <200 mV during Wi-Fi or ADC activity.

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5. I2C Bus Idle (before firmware)

Scope: Ch1 = I2C_SDA_3V3 (IO0 side), Ch2 = I2C_SCL_3V3 (IO1 side), DC coupling, 1 V/div, 1 ms/div.

5.1 3.3V bus (ESP32 and INA226)

- SDA and SCL idle at **3.3 V**, clean edges, no glitches.

5.2 5V bus (ADS1115)

- Move Ch1 to I2C_SDA_5V and Ch2 to I2C_SCL_5V.
- Idle at **5 V**. A stuck low means a shorted line or a failed Q1/Q2.

5.3 Rise time

- 100 ns/div on a rising edge, once firmware is driving the bus.
- Expect a rise of about 200 to 500 ns (RC with R18/R20 or R19/R21).

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6. Thermistor Front-End

6.1 Empty jacks (DC, DMM)

- With no plug, the signal net (TP10 PIT_IN, TP11 MEAT1_IN, TP13 MEAT2_IN, TP14 MEAT3_IN) reads **~0 V**, because the jack wire shorts it to GND.
- With a probe inserted, the signal net rises to the bias divider voltage (section 6.2).
- A signal net near +5 V with no plug means the jack wire or the GND connection is missing. Stop and check.

6.2 Known-resistance test

Insert a plug with a test resistor from the signal pin to GND, and measure the node. For the 120k bias to +5V:

Test resistor	Expected node voltage
10 kΩ	~0.38 V
47 kΩ	~1.47 V
100 kΩ	~2.27 V
1 MΩ	~4.46 V

Divider: V = 5 V × R / (R + 120 kΩ). Check the measured values against these. This test uses a plug and resistor, not a thermistor.

6.3 Filter noise (scope)

- TP10 on Ch1, AC coupling, 10 mV/div to 20 mV/div, 10 ms/div.
- Expect <50 mV peak-to-peak with no plug or a stable resistor. Look for 50 or 60 Hz pickup.
- At 17.5 Hz cut-off, a 1 V step should settle in roughly 30 ms.

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7. ADS1115 (U6, 0x48) – before firmware PGA is coded

Do not interpret ADC readings until the firmware sets PGA and MUX. The power-on defaults are: - PGA = ±2.048 V full scale - MUX = AIN0 to AIN1 (differential)

With those defaults, a 2.27 V single-ended input will read saturated, and the differential reading will not match any channel. Use the DMM for the thermistor nodes until the config register is written.

7.1 Supply

- Ch1 on U6 VDD (pin 8), DC coupling, 1X probe, 50 mV/div, 500 µs/div.
- Expect 5 V with <100 mV ripple.

7.2 Address

- ADDR is tied to GND, so the device answers at **0x48**.

7.3 Once PGA is coded

- Single-ended readings per channel (AIN0 PIT, AIN1 MEAT1, AIN2 MEAT2, AIN3 MEAT3).
- Recheck the table in section 6.2 against the ADC codes.
- Set PGA to match the chosen full scale.

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8. INA226 (U2, 0x40) – current and bus voltage

INA226 is on the 3.3V bus. Its shunt sense inputs measure R8 (100 mΩ) in the +5V path.

8.1 Address and supply

- Device answers at **0x40** on the 3.3V bus.
- Check VS = +3V3 with the DMM.

8.2 Shunt drop (DMM, mV scale)

- Across R8 with the board drawing about 50 mA: expect ~**5 mV** (50 mA × 0.1 Ω).
- Above 20 mV at idle: investigate current draw.

8.3 Bus voltage

- Vbus pin 8 on +5V. Expect the same reading as TP5 once firmware reads it.

8.4 Kelvin check

- Measure across R8 at the sense pads, not at the trace near the power connector. Use a Kelvin probe.

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9. ESP32 Boot and Programming

9.1 Boot mode

- Hold **SW2 (BOOT, IO9)**, tap **SW3 (RESET, EN)**, then release BOOT. This selects the ROM download mode.
- IO9 is strapped high by R16 (10k). Pulling it low at reset enters download mode.
- The display reset (DISP_RST) shares this net, so the display may reset during boot. Expected.

9.2 USB

- Connect a computer to the USB-C input only with a data-capable cable. A power bank gives power only.
- The ESP32 enumerates over IO18/IO19 (native USB). Check the port appears on the host.
- If the board enumerates only as a serial device, the bootloader is running. If it does not enumerate, check U5 and the D+/D− traces.

9.3 Debug UART

- Debug header (emergency debug and programming only, not connected in normal use): Pin 1 GND, Pin 2 DEBUG_TX/LED (IO21), Pin 3 DEBUG_RX (IO20), Pin 4 3V3.
- Use a **3.3V** USB-serial adapter only. Do not connect a 5V adapter to these pins.
- 115200 baud. Expect the boot ROM and firmware banner.

9.4 LED3 behaviour

- LED3 is on IO21 (TXD). It will flicker or glow during serial output, not steadily blink from a GPIO.

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10. Firmware I2C Scan

Minimal sketch, in your own code: 1. Initialise UART at 115200 on IO20/IO21. 2. Print "cloudSmoker2 alive". 3. Initialise I2C on IO0 (SDA), IO1 (SCL), 100 kHz. 4. Scan 0x03 to 0x77. 5. Print each device found.

Expected results

- **0x40 present:** INA226 (3.3V bus).
- **0x48 present:** ADS1115 (5V bus, through the level shifter).
- No other addresses.

If a device is missing

- **0x48 missing:** check +5V at U6 VDD, I2C_SDA_5V and I2C_SCL_5V idle high, Q1 and Q2 conduction, then the level shifter.
- **0x40 missing:** check +3V3 at U2 VS, I2C_SDA_3V3 and I2C_SCL_3V3 idle high, then the INA226 solder joints.
- Both missing: check that IO0 and IO1 are driving the bus and that pull-ups are present.

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11. Subsystems (with firmware)

11.1 Display (GC9A01 8-pin module on JST-XH headers J3 and J6)

- Display is powered from +5V and takes 3.3V logic (signal-safe).
- SPI on IO6 (SCK), IO7 (MOSI), IO8 (CS), DC on IO4, backlight PWM on IO5.
- Reset shares IO9.
- Expect the GC9A01 to initialise and show an image. If no image: check +5V at the display header, SPI_SCK on TP7, and that the backlight transistor is conducting.

11.2 Encoder (EC11 module on J8 male and J7 female, both 5-pin JST-XH)

- Pin 1 GND, pin 2 ENC_DT, pin 3 ENC_CLK, pin 4 ENC_SW, pin 5 3V3.
- Encoder runs from 3V3 with 3.3V logic, so no level shifting.
- ENC_CLK on IO3, ENC_DT on IO2, ENC_SW on IO10.
- Rotate and press: confirm edges in the firmware log.

11.3 Thermistor jacks (off-board, panel-mounted)

- Plug a known test resistor first (section 6.2), then a real probe.
- Confirm ADS1115 codes move with the probe temperature.

11.4 INA226 readback

- Bus voltage ~5 V, current ~50 mA at idle.

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12. Current and Thermal

- Idle (no display, no Wi-Fi): roughly 20 to 60 mA.
- Display on: plus about 20 to 40 mA.
- Wi-Fi active: plus about 50 to 100 mA.
- Anything above 300 mA sustained: investigate.
- After 5 minutes: LDO U1 warm, ESP32 warm, ADC cool. Above about 60 °C on any part: power off.

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13. Go / No-Go

Before first power

- [ ] +5V to GND >10 kΩ, +3V3 to GND >5 kΩ
- [ ] R18, R20 ≈ 5.1k; R19, R21 ≈ 10k (or 9.1k)
- [ ] Q1 and Q2 present, no shorts on gate, source, drain
- [ ] Jack pins 1, 2, 3 continuous with no plug; signal net to GND ~0 Ω
- [ ] No solder bridges on QFN and fine-pitch parts

After power-on

- [ ] PSU current 20 to 60 mA, no ramp spike above 150 mA
- [ ] TP5 4.9 to 5.1 V, TP3 3.2 to 3.4 V
- [ ] TP6 (EN) ~3.3 V
- [ ] LED1 and LED2 on
- [ ] No hot parts

After firmware scan

- [ ] 0x40 (INA226) and 0x48 (ADS1115) found
- [ ] No extra addresses
- [ ] UART banner printed at 115200

After PGA coded

- [ ] Empty jacks: signal net ~0 V
- [ ] Test resistors match section 6.2 table
- [ ] INA226 current ~50 mA at idle, shunt drop ~5 mV

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14. Scope Reference

Signal	Location	Expected	Red flag
+5V	TP5	4.9–5.1 V, <100 mV p-p	<4.8 V, >5.3 V, >200 mV p-p
+3V3	TP3	3.2–3.4 V, <50 mV p-p	<3.0 V, >3.6 V, >100 mV p-p
RESET / EN	TP6	~3.3 V	<1 V (stuck low)
IO9 BOOT	TP8	~3.3 V, high at boot	stuck low
I2C_SDA_3V3	IO0 (pin 18)	3.3 V idle	stuck low, ringing
I2C_SCL_3V3	IO1 (pin 17)	3.3 V idle	stuck low, ringing
I2C_SDA_5V	ADS1115 SDA	5 V idle	stuck low, ringing
I2C_SCL_5V	ADS1115 SCL	5 V idle	stuck low, ringing
PIT_IN	TP10	~0 V empty (jack wire to GND); divider with probe	near +5 V with no plug (jack wire missing)
MEAT1_IN	TP11	same as PIT_IN	same
MEAT2_IN	TP13	same as PIT_IN	same
MEAT3_IN	TP14	same as PIT_IN	same
Shunt R8	across 100 mΩ	~5 mV at 50 mA	>20 mV at idle

15. Troubleshooting

Symptom	Likely cause	Check
PSU current >150 mA on ramp	Short on +5V or +3V3	Power off, re-measure rail-to-GND
+3V3 reads 0 V	LDO input short or dead LDO	TP5 level, U1 IN/OUT pins
TP6 reads 0 V	EN shorted to GND	Check SW3 and R21
I2C_SDA_5V stuck low	Q1 shorted or 5V-side short	Q1 source/drain, trace to U6
I2C_SDA_3V3 stuck low	IO0 or R18 short	R18 value, IO0 to GND
0x48 missing	5V side not powered or level shifter open	+5V at U6 VDD, Q1/Q2
0x40 missing	INA226 not powered or solder	+3V3 at U2 VS, solder joints
Signal net near +5 V with no plug	Jack wire or PCB GND connection missing	Jack pins 1–3 continuity, signal to GND at PCB
Signal net stays at 0 V with probe in	Probe not making the internal break, or signal open	Jack pin 2 to probe tip, PIT_IN trace
Thermistor reads not matching table	Wrong bias, filter, or unset PGA	Recheck 6.2 table and section 7
No display image	SPI, backlight, or reset	SPI on TP7, +3V3 at header, IO9
No UART output	Adapter level, wrong pins	3.3V adapter, debug header pins 2 and 3 (TX, RX)